University of Cape Town
Department of Computer Science

Computer Science CSC3003S
Concession Test, October 2022

Answer all questions.
Marks per question are shown in brackets.

Time: 45 minutes **Marks: 35**

Section 1: ASD 1 [9]

Question 1. **[3]**

One of the values of the Agile Manifesto starts with "Individuals and Interactions" as the "item on the left". Give the "item on the right" that this is contrasted with (e.g. working software over comprehensive documentation) explain the value and its importance in 1-2 sentences, and give an example of an agile principle which captures some aspect of the value.

ANSWER:

1 point for the correct item on the right: Processes and Tools

1 point for naming an appropriate agile principle

- The most efficient and effective method of conveying information is face-to-face conversation
- Continuous attention to technical excellence and good design
- The best arch, req, and designs emerge from self-organizing teams
- The team reflects on how to becom emroe effective and adjusts its behavior accordingly
- Build projects around motivated individuals. Give them the support they need. Trust them.

1 point for clearly explaining the value and why it is important, it has to do more than restate the value

Comparing to waterfall, in which processes and tools drive everything, in agile, people drive the development, and valuing human interaction and individual agency is important.

Question 2. **[6]**

a. All involved in Scrum are either chickens or pigs. Explain. [1]

ANSWER:

Chickens are spectators – interested and informed, but spectators and nor responsible. Pigs are players, committed accountable and responsible. In scrum you want to minimize chickens and maximize ownership of pigs.

b. Name the three SCRUM roles, state their responsibilities, and whether each is a chicken or pig. [5]

ANSWER:

2 points for naming all three correctly, 1 point for 2 correct, 0.5 for naming just one.

a. Product Owner: represents the interest of everyone with a stake in the project and its resulting system (probably a chicken, but ideally a pig) [0.5] Manages the product backlog and prioritizes it [0.5]
b. Team: responsible for developing functionality and deciding how to turn product backlog into functionality in a given iteration [0.5]; responsible for managing their own work [0.5] collectively responsible for project as a whole
c. Scrum Master: teaches scrum to everyone in project

Section 2: ASD 2 [8]

Question 3. **[3]**

Briefly explain the following SOA concepts:

a. Software as a Service (SaaS)

ANSWER:

SaaS means offering software functionality to users remotely over the web, rather than through applications installed on a user's computer [1.5].

b. Service Oriented Systems (SOS)

ANSWER:

SOS are systems that are implemented using reusable service components and that are accessed by other programs, rather than directly by users [1.5]

Question 4. **[5]**

Chain of responsibility, visitor, strategy, and command are examples of the behavioural category of design patterns. Briefly explain each of these examples.

ANSWER:

(i) Chain of responsibility:
Used to achieve loose coupling in software design where a request from the client is passed to a chain of objects to process them [1.5]
(ii) Visitor:
This pattern is used when we have to perform an operation on a group of similar kinds of objects [1]
(iii) Strategy:
Is used when we have multiple algorithms for a specific task and client decides the actual implementation to be used at runtime [1.5]
(iv) Command:
This pattern is used to implement lose coupling in a request-response model [1]

Section 3: Theory of Algorithms [9]

Question 5. **[2]**

You are given a recurrence relation for an algorithm:

$$T(n) = 3T\left(\frac{n}{4}\right) + n^3 + 1, for\ n > 1, and\ T(1) = 5$$

Solve this using the Master Theorem.

ANSWER:

a = 3, b = 4, c = 5, d = 3 1 mark

→ a = 3 < b^d = 4^3 0.5 mark

→ T(n) ∈ $\Theta(n^3)$ 0.5 mark

Question 6. **[3]**

Given the text *t* = BIB_A_BITT

And the pattern *p* = BIT

Construct the shift table for Horspool's algorithm for the alphabet {A-Z, _}, then run the matching algorithm, showing the steps/shifts taken when the pattern *p* is run against the text *t*

ANSWER:

Table:

B	I	Other letters, _
2	1	3

½ mark for B

½ mark for I

½ mark for other alphabet characters

Apply

```
BIB_A_BITT
BIT
  BIT (2 shift)                    ½ mark {reason: T(B) = 2}
     BIT  (3 shifts)                 ½ mark {reason: T(A) = 3}
      BIT (1 shifts = match!)    ½ mark {reason: T(I) = 1}
NOTE: reason need not be written for the mark, shifts must be
correct though
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Question 7. **[3]**

You are given the following knapsack problem solution, for up to 5 items (with indicated weights and values) and a knapsack of maximum capacity 6.

		capacity j						
	i	0	1	2	3	4	5	6
	0	0	0	0	0	0	0	0
$w_1 = 3, v_1 = 25$	1	0	0	0	25	25	25	25
$w_2 = 2, v_2 = 20$	2	0	0	20	25	25	45	45
$w_3 = 1, v_3 = 15$	3	0	15	20	35	40	45	60
$w_4 = 4, v_4 = 40$	4	0	15	20	35	40	55	60
$w_5 = 5, v_5 = 50$	5	0	15	20	35	40	55	65

Use **back-tracing** to determine the optimal set of items included in a knapsack of up to **3** items and a capacity of **6**. Show each step.

ANSWER:

We need to start with F(3,6) = 60, ½ mark
Then, check if F(3,6) == F(2,6) → NO. This means **item 3** is in the set, ½ mark
We then move to cell i = 2, j = 6-w_3 = 5 ½ mark
Then we check if F(2,5) == F(1,5) → NO. **So item 2** is in the set ½ mark
We then move to cell i = 1, j = 5-w_2 = 3 ½ mark
Then we check if F(1,3) == F(0,3) →NO, so **item 1** is in set ½ mark
We then move to cell i = 0, j = 3-w_1 = 0 -> done

Question 8. **[1]**

Give an example of where the "greedy change making" algorithm fails to give the least amount of change for a desired amount, assuming currency denomination = {1, 15, 25}.

ANSWER:

many – example Value = 30, gives 30-25, then 5 X 1 coins for 6 coins, instead of 2 x 15 coin. (any example where the change is not minimal, based on common sense!)

Section 4: Theory of Computation [9]

Question 9. **[1]**

Give a regular expression for the language of binary numbers i.e. strings comprising any number of 0s and 1s in any order.

ANSWER:

(0|1)*

Question 10. **[4]**

Draw a transition diagram for a Turing Machine that converts an input binary string into its one's complement form – i.e. every 1 is replaced by a 0 and every 0 is replaced by a 1. Assume that the tape head starts at the cell where the input binary string starts. The computation must end with the tape head at the first blank cell encountered.

ANSWER:

1 mark for correctly showing initial and halt state

1 mark for correctly showing nodes, edges and edge labels

2 marks (all or nothing here) if correct for any input (including blank tape i.e. no input).

Machine should have 3 states, I'll call them F (first, the initial state), D (done a digit) and H (halt).

Arc from F to H labelled blank/blank

2 arcs from F to D: one labelled 0/1 and the other labelled 1/0

2 arcs from D to F: one labelled 0/R and the other labelled 1/R

Question 11. **[2]**

Give ***any one*** entry that would appear in the state table for your Turing Machine above.

ANSWER:

1 mark for columns; 1 mark for values in the example entry exactly matching any arc in their TM (whether or not their TM does one's complement correctly or not)

Columns are:

current state; current symbol; next state; action

action is 1 of these, can indicate which in any way:

L/move Left or R/move Right or O (write 0) or 1 (write 1) or blank (write blank)

e.g. the arc from F to H in my example would have entry as below:

F, blank, H, blank

etc.

Question 12. **[2]**

Briefly state the difference between an acceptable language and a decidable language.

ANSWER:

Same as in original test.

Acceptable: there is an algorithm that can halt and say Yes if and only if it is given a string in the language

Decidable: there is an algorithm that will halt and say Yes if it is given a string in the language and will halt and say No if it is given a string that is not in the language

Appendix A – Master Theorem

If we have a recurrence of this form: $T(n) = aT(n/b) + f(n)$, $f(n) \in \Theta(n^d)$ and $T(1) = c$ then

$T(n) \in \Theta(n^d)$ if $a < b^d$

$T(n) \in \Theta(n^d \log n)$ if $a = b^d$

$T(n) \in \Theta(n^{\log_b a})$ if $a > b^d$